Homework — 1.4.2 Data Structures

Name: ____ Date: _ ***Mark:*** ___ / 20

OCR H446 — set after the 1.4.2 lesson. SHOW ALL WORKING.

Part A — This week's topic (~14 marks)

1. A **stack** is initially empty. The following operations are carried out in order. Show the stack after each operation, then give the values popped (in order) and the final contents. **[4]** *(AO2)*

`push(8)`, `push(3)`, `push(6)`, `pop()`, `pop()`, `push(1)`

Working space (show the stack after each operation):

Values popped (in order): _ ***Final stack (top → bottom):*** _

2. Explain why a **circular queue** is often preferred over a simple linear queue that uses front and rear pointers in a fixed array. Refer to how the rear pointer is updated. **[3]** *(AO1)*

Working space:

3. The following numbers are inserted, in this order, into an empty **binary search tree**: **40, 60, 25, 35, 10, 50**. Draw the resulting tree. **[3]** *(AO2)*

Working space:

4. Give the **in-order** traversal of the binary search tree you drew in Question 3, and state what is special about the sequence produced. **[2]** *(AO2)*

Working space:

5. A hash table of size **10** uses the hash function `index = key MOD 10`. The keys **23, 57, 33** are inserted in that order. State the index each key hashes to, identify any **collision**, and name **one** method of resolving it. **[2]** *(AO2)*

Working space:

Part B — Retrieval: keep it fresh (~6 marks)

6. Explain why an operating system uses a **LIFO stack** to handle **nested interrupts**. *(1.2)* **[2]** *(AO1)*

Working space:

7. State the role of the **control bus** in a processor. *(1.1)* **[1]** *(AO1)*

Working space:

8. A relational database links two tables using a **foreign key**. State what a foreign key is, and explain what **referential integrity** ensures. *(1.3)* **[3]** *(AO1)*

Working space: